

# Licensed Electrician's Theory (LET) Assessment

## Sample Paper Three Marking guide (2021)



### AS/NZS 3000 Wiring Rules

#### Q.1

Point of entry (2 marks) Clause 1.4.94 (2 marks)

#### Q.2

1. Socket outlets shall not accept plugs of other voltages, AND 2. Socket outlets shall not have a contact for a protective earthing conductor. (1 mark for each part for a total of 2 marks) Clause 7.5.10 (b) (c) (2 marks)

#### Q.3

2 m (2 marks) (Deduct 1 mark for no or incorrect units)

Clause 4.7.1 NOTE (2 marks)

#### Q.4

Where installed immediately below a ceiling (2 marks)

Clause 4.5.1.3 Exception (2 marks)

### AS/NZS 3012 Construction & Demolition Sites.

#### Q.5

The requirement out of reach need not apply in stairwells, lift shafts, and service shafts that support prevents sagging of conductors shall be provided (2 marks)

Clause 2.7.6 (c) Exception (2 marks)

#### Q.6

160 lx (Deduct 1 mark for no or incorrect units) (2 marks)

Clause 2.7.1 (2 marks)

### Electrical Safety (General) Regulations 2019

#### Q.7

a) at the main switchboard; b) at an earth bar or link within a substation; or c) made through an earthing conductor or terminal provided by the electricity supplier (1 mark for each, for a total of 2 marks)

Reg 208 (1) (a) (b) (c) (2 marks)

## Electrical Shock Survival

### Q.8

Place the casualty on his/her back (2 marks)

Tilt the head back and raise their chin forward (2 marks)

## Cable Selection

### Q.9

Table 3(3) Item 4 (1 mark)

Table 14 Col 23 (2 marks, one for table and one for column)

Derating factor Table 25(2) Column 2 - 0.81 (1 mark)

Derating factor Table 28(1) Column - 0.90 (1 mark)

Part (i) Answer: 250.78A (2 marks)

Part (ii) B3.2.2.3 - 0.9

Answer: 225.78A (1 mark)

(Deduct 1 mark for no or incorrect units on final answers. Only deduct one mark regardless of number of missing units)

## DC Circuits

### Q.10

(i)  $I_c = 1.0A$  (2 marks)

(ii)  $V_a = 50V$  (2 marks)

(iii)  $P_b = 140W$  (2 marks)

(Deduct 1 Mark for no or incorrect units)

## Maximum Demand

### Q.11

Table C2 Column 3 (1 mark)

**Domestic Residence.**

| Equipment                                                         | Load Group    | Calculation                                                                                               | Maximum Demand   |
|-------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------|------------------|
| 4 ccts each of 10 - 10A Double socket outlets<br>40 Points total  | (b)(i)        | $1000 + (39 \times 750) = 30250$<br>$30250/230 = 131.52A$                                                 | 131.52A (1 mark) |
| 2 ccts each of 2 - 15A socket outlets<br>4 Points total           | (b)(iii)      | Full Connected Load of the highest<br>+ 75% of the remainder<br>$15 + (3 \times 15 \times 0.75) = 48.75A$ | 48.75A (2 mark)  |
| 3 ccts each of 8 – incandescent lights<br>24 lights               | (a)           | $24 \times 60 = 1440/230 = 6.26A$                                                                         | 6.26A (1 mark)   |
| 1 cct – Petrol pump (4A motor)                                    | (f)           | Full Load Current of the highest motor = 4A                                                               | 4A (1 mark)      |
| 1 cct supplying an arc welder with a primary current rated at 25A | (h)<br>C2.5.2 | C2.5.2.2 (a) Maximum demand of the welder 100% of the primary rated current = 25A                         | 25A (1 marks)    |
|                                                                   |               | Total Maximum Demand                                                                                      | 215.53A (1 mark) |

(Deduct 1 Mark for no or incorrect units on total, deduct 1 mark for no or incorrect load groups)

## Voltage Drop

### Q.12

**Consumer Mains**

Table 42 Column 6 (1 mark)

Vc 1.54 (1 mark)

Vd 4.50V (1 mark)

**Sub-mains**

Table 41 Column 6 (1 mark)

Vc 3.86 (1 mark)

Vd 4.86V (1 mark)

**Final Sub-circuit**

Table 41 Column 4 (1 mark)

Vc 6.18 (1 mark)

Vd 11.12V (1 mark)

Total Voltage Drop =  $4.50 + 4.86 + 11.12 = 20.48V$  (1 mark)

(Deduct 1 Mark for no or incorrect units on total. Deduct 1 mark for no or incorrect table number/s)

LET Sample Paper Marking Guide 2021

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## Overload & Short Circuit Calculations

### Q.13

Overcurrent divided by MCB current rating = 4 (1 mark)

Minimum Time = Accept 2 - 2.5 seconds (1 mark)

Maximum Time = Accept 7 - 9 seconds (1 mark)

(Deduct 1 mark for no or incorrect time unit)

### Q.14

#### Transformer impedance

230/13,900 (2 marks)

0.01655Ω (1 mark) Answer must be to 5 decimal places.

#### Main switchboard prospective fault

230/ (0.01655 +0.0045) (2 marks)

10,926.37A (1 mark)

#### Distribution board prospective fault

230/ (0.01655 +0.0045+ 0.02) (2 marks)

5602.92A (1 mark)

(Deduct 1 Mark for no or incorrect units in final answer)

## Residual Current Devices

### Q.15

16A (1 Mark)

AS/NZS 3000 Clause 2.6.2.1 (b) (2 marks)

## Motor and Starters

### Q.16

C (2 Marks)

## AS/NZS 4836:2023

### Q.17

Any two of the following: a) Restricted areas in and around switchboards; b) Ceiling and roof spaces; c) Spaces under floors; d) Ladders, scaffolds, or elevated work platforms; e) trenches; f) pits or tunnels; g) Confined spaces. (1 mark for each for a total of 2 marks) Clause number: 4.3.5 (a) – (g) (2 marks)

## Installation Defects- Non Domestic

### Q.18

(2 marks for correct defect one mark for the correct clause)

(Only accept the first 5 defects candidate has listed)

1. – Submain not of sufficient current carrying capacity for the DB load 3.4.1 T14 C2 147A
2. – Telecommunications protective earth is less than the required minimum of 6mm<sup>2</sup> 5.6.2.7 (iv)
3. – Crane final sub circuit conductors are not of sufficient current carrying capacity for the load 3.4.1 T7 C15 35A
4. – Oven final sub circuit conductors are not of sufficient current carrying capacity for the load 3.4.1 T19 C4 96A
5. – The Oven final sub circuit conductors have no overcurrent protection device 2.5.1.3
6. – MEN link is undersized 5.3.5.2
7. – Oven main switch is not marked “Main Switch” 2.3.3.5(a)
8. - Location of Earth Electrode not indicated at Main Switchboard 5.3.6.4
9. - Main Earthing conductor connection to Earth Electrode label (tag) does not state “Warning: Main Electrical Earthing Conductor – Do Not Disconnect” 5.5.1.3
10. -Neutral Bar not identified as such 2.10.5.4 ??